Short-mode white noise in a radiation universe: spatial curvature and the measured temperature

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Verdict

Spatial curvature. Yes. Short-wavelength modes alone generate a finite white-noise term in the comoving curvature perturbation: $P_{\mathcal{R}}(K) \to \epsilon^4 A_0(t) > 0$. There is no $1/K^2$ enhancement and no suppression by powers of K. The coefficient is explicit, Eqs. (7)–(9). Its direction-dependent part is a projection of the shear driven by the short modes' effective anisotropic stress, and it grows as $\ln(qt)$ once they are inside the horizon. The statement depends on the threading in the definition of $\mathcal{R}$: the conserved volume variable ζ has no response at all, to all orders.

Measured temperature. Yes, it survives, but not as curvature. The full normalised map has a nonzero K^0 quadrupole, Eq. (13). It is checked against an independent nonlinear calculation to 5×10^{-6} or better. Its generated part is $-\frac{1}{2}\hat{n}^i\hat{n}^j\Delta S_{ij}$, the time *difference* of the tensor whose $\hat{K}\hat{K}$ projection appears in $\mathcal{R}$, and it grows as $-8\ln(t_o/t_\star)$. Direct short-mode terms of the same order (emission surface, observer boost, ray) are present and bounded, except for an emission term growing like $qt_\star$ with zero mean; they do not cancel the logarithm. The absence of a logarithm in the ray term is established numerically, not by proof.

This is outcome 1 of the question: an explicit nonzero coefficient. A white-noise $P(K)$ does not imply a flat C_ℓ, and no C_ℓ is computed here.

1 Setup and the definition of $\mathcal{R}$

I use the main report's conventions. For an irrotational radiation fluid, $Tu_\mu = -\mathcal{T}\partial_\mu t$ defines a potential t. In coordinates (t, X^i), with X^i comoving material labels,

$$g = N^2(-\mathrm{d}t^2 + \gamma_{ij}\mathrm{d}X^i\mathrm{d}X^j), \qquad N = a_1 t e^\nu = \mathcal{T}/T, \qquad \gamma = e^{2\zeta}e^H, \quad \operatorname{tr} H = 0. \tag{1}$$

The primordial state is the regular solution with $\zeta = \epsilon\zeta_1(X)$ exactly and ζ_1 Gaussian, with $\gamma \to e^{2\zeta}\delta$ and $\nu, H \to 0$ as $t \to 0$. The spectrum is smooth with support in $q_{\min} < q < q_{\max}$.

Definition. *Slicing:* level sets of t, the hypersurfaces orthogonal to u. *Threading:* on each slice, the chart x in which the induced metric takes the form

$$h_{ij} = a^2 e^{2\mathcal{R}}(e^\chi)_{ij}, \qquad \operatorname{tr}\chi = 0, \quad \partial_j\chi_{ij} = 0, \qquad a = a_1 t. \tag{2}$$

Geometric meaning: $\mathcal{R}$ is the conformal factor of the comoving slice in the chart where the unimodular part is transverse, so χ does not enter the scalar curvature linearly, and $a^2\,{}^{(3)}R = -4\nabla^2\mathcal{R} + 8\mathcal{R}\nabla^2\mathcal{R} - 2(\partial\mathcal{R})^2 + \ldots$. It is a nonlocal functional of the 3-geometry; the chart exists perturbatively and is unique at nonzero wavenumber. With $X = x + \xi(x,t)$ and $J = \det \partial X/\partial x$, the determinant gives the exact relation

$$\mathcal{R}(x,t) = \zeta(X) + \nu(X,t) + \tfrac{1}{3}\ln J. \tag{3}$$

As $t \to 0$, $\mathcal{R} = \zeta$ exactly. Through second order, with $\xi_1 = \partial\beta_1$, $m = \partial\partial\beta_1$, $\psi = \zeta + \nu$, and first-order threading fixed by $\chi_1 = 0$,

$$\mathcal{R}_2 = \psi_2 + \xi_1\cdot\partial\psi_1 - \tfrac{1}{6}\operatorname{tr} m^2 - \tfrac{1}{4}\hat{K}_i\hat{K}_j S_{ij}, \qquad S = H_2 + \xi_1\cdot\partial H_1 - \mathrm{TF}\, m^2. \tag{4}$$

This form is verified off shell, for generic momenta and an arbitrary trace-free H_2, by matching the scalar curvature in the two charts (`check_chart_R.py`).

At first order, $x = qt/\sqrt{3}$, $\nu_1 = \cos x - j_0(x)$, $j_0 = \sin x/x$, and $H_1 = h(x)\,\mathrm{TF}(\hat{q}\hat{q})\zeta_1$ with $h = 6(1 + \cos x - 2j_0)$. This gives $\mathcal{R}_1 = j_0(x)\zeta_1$, which starts at 1 and decays inside the horizon.

2 First route: the decisive calculation

For the seed $\zeta_1 = A\cos(q\cdot x) + B\cos[(q-K)\cdot x]$, the output at K is $(AB/2)\cos(K\cdot x)\,K_2$, where $K_2 = 2F_2$ is the coefficient of $z_1 z_2$ for the pair $(q, K-q)$ with $z_1 z_2 = AB/4$.

Why the limit is finite and direction dependent. The trace-free equation has an entire Green function in the Cartesian components of K (main report, "Projector-free Cartesian solution"), so H_2 and ν_2 are continuous at $K = 0$. Hence $H_2 \to a(x)\,\mathrm{TF}(\hat{q}\hat{q})$ and $\nu_2 \to n_2(x)$. But the projector in (4) remains: $\hat{K}\hat{K}\cdot\mathrm{TF}(\hat{q}\hat{q}) = \mu^2 - \frac{1}{3}$, with $\mu = \hat{K}\cdot\hat{q}$. The direction is not averaged away, and the limit is not analytic in K.

The $K = 0$ pair fields in closed form. With ζ along one axis the exact equations reduce to 1+1 dimensions. The pair sources integrate by hand; all Si terms cancel:

$$n_2 = -\cos 2x - 6\cos x + 6j_0 + 3\frac{\sin 2x}{x} - 5j_0^2, \tag{5}$$

$$a = -78 - 36\cos x - 6\cos 2x + 72j_0 + 36\frac{\sin 2x}{x} - 24j_0^2 + 24\,\mathrm{Cin}(2x). \tag{6}$$

Substitution into (T), (N), (M) and (Ham) gives exact zeros, and mutated coefficients fail. Also $a = 2x^2 + O(x^4)$, $n_2 = \frac{5}{3}x^2 + O(x^4)$, which match the exact gradient data. Assembling (4), the terms linear in $\cos x$ cancel and

$$\boxed{F_2^{\mathcal{R}}(q, K-q) \xrightarrow{K\to 0} \tfrac{1}{2}I(x) - \tfrac{1}{8}\Big(\mu^2 - \tfrac{1}{3}\Big)W(x), \qquad \begin{aligned} I &= 4 + \cos^2 x - 5j_0^2, \\ W &= 24\,\mathrm{Cin}(2x) + 6\cos^2 x - 54 + 48j_0^2. \end{aligned}} \tag{7}$$

So the result is K^0: no pole and no vanishing constant. Writing $F_2 \to \alpha + \beta P_2(\mu)$ with $\alpha = I/2$ and $\beta = -W/12$,

$$\alpha = \frac{x^2}{3},\quad \beta = -\frac{x^2}{6} \quad (x \ll 1); \qquad \alpha \to 2 + \tfrac{1}{2}\cos^2 x,\quad \beta \to -2\big[\ln 2x + \gamma_E - \tfrac{9}{4} + \tfrac{1}{4}\cos^2 x\big] \quad (x \gg 1). \tag{8}$$

This is valid for $K \ll q$ and $Kt \ll 1$ at fixed t, including $qt \gg 1$.

The power spectrum. The angular average of P_2 over $\hat{q}$ vanishes, but that of $|F_2|^2$ does not. With $\langle P_2^2\rangle = \frac{1}{5}$,

$$\boxed{P_{22}(K\to 0) = A_0(t) = \frac{1}{\pi^2}\int_0^\infty q^2\mathrm{d}q\, P_{\zeta_1}^2(q)\Big[\alpha^2 + \frac{\beta^2}{5}\Big] > 0.} \tag{9}$$

Early on, $A_0 \to \frac{7t^4}{540\pi^2}\int q^6 P^2 \mathrm{d}q$. Late, the bracket is dominated by $\frac{4}{5}\ln^2(qt)$. The quantity A_0 is dimensional power: the dimensionless spectrum is $K^3 A_0/2\pi^2$. Although the kernel is non-analytic in $\hat{K}$, the spectrum is isotropic, so P_{22} is constant as $K \to 0$. The exact kernels at finite K converge to (7) at all tested angles, including $|q_1| \neq |q_2|$. A direct quadrature for a smooth compact bump centred at q_0 with half-width $0.3q_0$ gives $P_{22}(K)/A_0 = 0.8589,\ 0.9565,\ 0.9867$ at $K/q_0 = 0.2,\ 0.1,\ 0.05$ for $q_0 t/\sqrt{3} = 1$, and $0.7685,\ 0.9327,\ 0.9782$ for $q_0 t/\sqrt{3} = 4$. These approach 1; the deficit shrinks by a factor of about 3.3 per halving of K, and I do not claim the exact power of the correction.

Why P_{13} is absent. $P_{13}(K) = 2\,\mathrm{Re}\langle X_1(K) X_3^*(K)\rangle$, and $X_1(K) \propto \zeta_1(K) = 0$ for $K < q_{\min}$, so it vanishes identically, as does P_{12}. This does not carry over to C_ℓ: $a_{\ell m}^{(1)}$ projects band modes onto ℓ through Bessel functions at finite distance, so $C_\ell^{(13)} \neq 0$ (main report).

Mechanism and physical check. At $K \to 0$ the exact equation is $a'' + 2a'/t = -S_T$ (primes in t, $q = 1$), with

$$S_T = -12\cos x - 8\cos 2x + \frac{24\sin 2x}{x} - \frac{48\sin^2 x}{x^2} + \frac{16}{x^2} - \frac{16\sin 2x}{x^3} + \frac{16\sin^2 x}{x^4}.$$

Its late-time mean is $-8/x^2$. Identifying $a'' + 2a'/t = (6/t^2)\Pi/\bar{\rho}$ gives $\Pi_{ij}/\bar{\rho} = 4\,\mathrm{TF}(\hat{q}\hat{q})$. This equals the acoustic stress $\frac{4}{3}\bar{\rho}\langle v_i v_j\rangle$ with $v = \sqrt{3}\sin x\,\hat{q}\,\zeta_1$, and it yields the $24\ln x$ in a. In the usual language $\dot{\mathcal{R}} \sim K^2\Pi_S$, the scalar stress potential obeys $\Pi_S \sim \hat{K}\hat{K}\Pi/K^2$, so the suppression factor K^2 cancels.

A single fixed tensor has helicity power $1:2:2$ (scalar:vector:tensor) over directions $\hat{K}$. The scalar white noise is therefore the same phenomenon as the known K^3 tail of induced gravitational waves [1, 2]. As an independent check, on the shell the $K \to 0$ limit of the Kohri–Terada form in the main report satisfies $F_T \to a/2$, agreeing to 3×10^{-11} for x up to 15.

Separation of contributions.

Primordial variable	None: $\mathcal{R} = \zeta$ at $t \to 0$, $\zeta_2 \equiv 0$. A local $\zeta \to \zeta + f\zeta^2$ would add a constant; here $f = 0$ by construction.
Constraint-required data at t_i	$F_2(x_i) = x_i^2/3 - \frac{1}{4}x_i^2(\mu^2 - \frac{1}{3}) + O(x_i^4)$. It is part of the same regular solution, vanishes as $t_i \to 0$, and is not an independent mode.
Measurement definition (threading)	Local products $h(1+\nu_1) - h^2/12 - \frac{1}{8}(\mu^2 - \frac{1}{3})h^2$ in K_2. They are bounded and contain the $\cos x$ terms that cancel against n_2 and a.
Evolution	n_2 (bounded, zero-mean oscillation at late times) and $-\frac{1}{4}(\mu^2 - \frac{1}{3})a$ (memory integral of the stress, with the logarithm).

Threading sensitivity. In material threading, the volume variable $\zeta + \nu$ has only n_2 at $K \to 0$: isotropic, local, with no logarithm. The isotropic part α of (7) can be removed by a local field redefinition. The $\hat{K}$-dependent part β cannot, since it is non-analytic in K, but it does vanish under a change of threading. So $\mathcal{R}$ alone does not establish a physical signal; the shear in the next sections does.

3 Second route: exact identities

Proposition 1 (All orders, for smooth irrotational flow)**.**

(a) Conservation. $\partial_t \det\gamma = 0$, *so* $\zeta_K(t) = \zeta_K(0)$ *for all* K. *Physically,* $e^{3\zeta}$ *is the entropy per unit label. Below the band* $\zeta_K = 0$ *exactly, with no singular term, no constant and no power of* K.

(b) Relation to $\mathcal{R}$. *Eq.* (3). *Thus (a) does not constrain* $\mathcal{R} - \zeta$*:* ν*, the composition with* X*, and* $\ln J$ *all contribute at* K.

(c) Transversality. *Define* $\mathcal{S}$ *by* $\ln(J^{-2/3}(\partial X)^T e^{H\circ X}\partial X) = \mathcal{S} + 2\,\mathrm{sym}\,\partial\xi - \frac{2}{3}\delta\,\partial\cdot\xi$. *Then* $\partial_i\partial_j\chi_{ij} = 0$ *gives, for* $K \neq 0$, $(\partial\cdot\xi)_K = -\frac{3}{4}\hat{K}_i\hat{K}_j\mathcal{S}_{ij,K}$. *Therefore*

$$\mathcal{R}_K = \mathcal{C}_K - \tfrac{1}{4}\hat{K}_i\hat{K}_j\mathcal{S}_{ij,K}, \qquad \mathcal{C} = \zeta\circ X + \nu\circ X + \tfrac{1}{3}(\ln J - \partial\cdot\xi). \tag{10}$$

(d) Shear balance, retaining all short gradients. *Since* $\partial_t\sqrt{\gamma} = 0$, *equation (T) gives*

$$\partial_t\big(N^2\sqrt{\gamma}\,S^i{}_j\big) = -N^2\sqrt{\gamma}\;\mathrm{TF}\left[R^i{}_j[\gamma] - 2D^iD_j\nu + 2D^i\nu D_j\nu\right], \quad S^i{}_j = \tfrac{1}{2}\gamma^{ik}\partial_t\gamma_{kj}. \tag{11}$$

The right side has a nonzero long-wavelength component made from short modes: this is the effective stress.

What this proves. If $\mathcal{S}$ and $\mathcal{C}$ have bounded spectra at small K, (10) excludes a singular enhancement. It does *not* exclude a constant. It identifies the constant as the local-composite spectrum of $\mathcal{C}$ plus the $\hat{K}\hat{K}$ projection of $\mathcal{S}$. Stronger suppression is false, because the $O(\epsilon^2)$ value of that constant is the nonzero (7). Note that $\mathcal{S}_2 = W\,\mathrm{TF}(\hat{q}\hat{q})$, the same W. Parts (a), (b) and (d) are exact for any smooth solution; (c) also needs the transverse chart to exist. Only the coefficient evaluation is order-by-order, and its boundedness hypothesis is verified perturbatively for compact spectra. No gradient expansion of the short modes and no soft input is used.

Order of limits and nonuniformity. The order here is $K \to 0$ at fixed (ϵ, t); then $\epsilon \to 0$; then, optionally, $qt \to \infty$. Two nonuniformities matter. (i) Fixed K with $t \to \infty$ leaves the regime $Kt \ll 1$, and the logarithm then saturates into propagating modes, which this calculation does not treat. (ii) Fixed ϵ with $qt \to \infty$ fails through acoustic steepening; a simple-wave estimate gives $\epsilon x_s \approx \frac{1}{2}$ for $\zeta = 2\epsilon \cos qy$. Exact plane-symmetric evolution agrees: the $K = 0$ average satisfies $\langle f\rangle/\epsilon^2 - a \approx 0.39$ ($\epsilon = 0.01$, $x = 10$), 3.39 ($\epsilon = 0.03$, $x = 10$), scaling as ϵ^2, and spectral steepening appears at $\epsilon x \approx 0.3$–0.4. After a shock, (a) fails with entropy production. The results therefore need $\epsilon qt \ll 1$.

4 The measured temperature

The exact redshift (main report) is $T_{\rm obs} = T_{\rm fluid}(o)\, e^{-I}$ with $I = \int \sigma_{ij} e^i e^j \mathrm{d}t$, and $T_{\rm fluid}(o)$ cancels on normalisation. Hence only shear can produce anisotropy; ν_2 and α enter only at higher order or not at all. The observer is the fluid line at fixed label X_o, with fixed proper time after $T = T_I$, and Θ is the normalised map, with the dipole removed at the end. Fields are Fourier analysed in X_o, since the observer is a label. In the chart x an extra bounded term $\xi_1 \cdot \partial\Theta_1$ would appear.

At $K \to 0$ the second-order map becomes a function of $\hat{n} \cdot \hat{q}$ alone, even in $\hat{n}$, with no dipole. For the seed,

$$\Theta(\hat{n}, X_o) \supset \frac{AB}{2} \cos(K \cdot X_o) \sum_{\ell \geq 2 \text{ even}} \mathcal{Q}_\ell\, P_\ell(\hat{n} \cdot \hat{q}). \tag{12}$$

The quadrupole coefficient is exactly

$$\boxed{\mathcal{Q}_2 = -\tfrac{1}{3}\big[W\big]_{x_\star}^{x_o} + \mathcal{Q}_2^{\rm em} + \mathcal{Q}_2^{\rm prod} + \mathcal{Q}_2^{\rm clk} + \mathcal{Q}_2^{\rm rem},} \tag{13}$$

where the pieces are defined as follows, with $\mathcal{Q}_2$ denoting the P_2 projection throughout.

- *Generated:* $-\frac{1}{2}\hat{n}^i\hat{n}^j[S_{ij}]_\star^o$, with $S = W\,\mathrm{TF}(\hat{q}\hat{q})$, the same tensor as in $\mathcal{R}$.
- *Emission surface* (closed form): $\mathcal{Q}_2^{\rm em} = -\frac{2}{3} t_\star h'(t_\star)\nu_1(t_\star) \to 2x_\star \sin 2x_\star$ for $x_\star \gg 1$. It is the first-order displacement of the surface times the local shear.
- *Normalised products* of first-order maps, $\mathsf{P}_0(J_a J_b)$: for large separation, $\to 2(\sin^2 x_o + \sin^2 x_\star)$, the squared Doppler terms at both ends.
- *Clock:* $-D(\partial_o J_a + \partial_o J_b)$, with $D = [-t_I^2\nu_1(t_I) - \int_{t_I}^{t_o} t\nu_1 \mathrm{d}t]/t_o$.
- *Remainder:* the ray deflection and direction terms $X_1 \cdot \partial H_1'$, $H_1'(\hat{n}, v_1)$, plus $\frac{1}{6}[h^2]_\star^o$. These are double time integrals, given explicitly in `map_k0.py`.

Leading coefficient. For $qt \ll 1$ everything is local, and

$$\mathcal{Q}_2 = -\tfrac{2}{9} q^2 (t_o^2 - t_\star^2) + O(q^4 t^4), \tag{14}$$

which is the $z_1 z_2$ part of the local quadrupole $-\frac{1}{6}\Delta(t^2)\, e^{-2\zeta}\,\mathrm{TF}[\partial\partial\zeta - \partial\zeta\partial\zeta]_{nn}$. It is a nonzero K^0 response, not suppressed relative to K^2 the way a linear long adiabatic mode is.

$x_\star$	x_o	$\mathcal{Q}_2$	$-\frac{1}{3}[W]$	rem	em	prod	clk
0.1	0.2	−0.019594	−0.0201	+0.0003	−0.0000	+0.0000	+0.0002
0.5	1	−0.319767	−0.5230	+0.1519	−0.0251	+0.0122	+0.0642
1	3	−4.268883	−2.7998	+1.3472	−0.2881	+0.1195	−2.6478
3	10	+0.292937	−8.1918	+0.1877	+6.8473	+0.8380	+0.6116
5	20	−15.3640	−10.1915	−1.8555	−7.3100	+3.4488	+0.5442
5	40	−20.7210	−16.5168	+1.7639	−7.3100	+1.5628	−0.2209
5	80	−24.8166	−21.0832	+0.3781	−7.3100	+3.0030	+0.1955
5	160	−33.9877	−28.5273	+0.9593	−7.3100	+0.9047	−0.0144

Table 1: The $K \to 0$ quadrupole coefficient in (13), $t_I = t_\star/2$. Printed values are evaluated from the closed fields and the report's map, by quadrature.

Short modes inside the horizon. Table 1 gives the evaluated pieces with $t_I = t_\star/2$. The large $\cos x_o$ oscillation in a is absent once the same combination W is used. For $x_o \gg x_\star \gg 1$,

$$\mathcal{Q}_2 \simeq -8 \ln \frac{x_o}{x_\star} + 4 \sin^2 x_o + 2x_\star \sin 2x_\star + O(1). \tag{15}$$

The coefficient -8 of the generated term is exact. The claim that the other terms add no logarithm is numerical: a regression of $\mathcal{Q}_2$ for $x_o - x_\star$ up to 110 gives $-8.17, -8.11, -7.90$ for $x_\star = 2, 5, 8$, and a $\cos 2x_o$ coefficient of $-2.01, -2.03, -1.79$ (predicted -2, the observer's second-order boost). It agrees with the physical argument that subhorizon potentials decay as x^{-2}, so ray rectification has no $1/t$ rate; that argument is not a proof.

Independent test of the observable. I solved the exact plane-symmetric Einstein–radiation system with $\zeta = 2\epsilon \cos qy$, integrated exact null rays with physical emission roots, the proper-time clock and the exact sky normalisation, and averaged over observer labels. This uses no perturbative map formula. Richardson extrapolation in ϵ reproduces $\mathcal{Q}_2$ with differences 3×10^{-8} to 5×10^{-6} for seven time pairs up to $(x_\star, x_o) = (5, 20)$. Omitting any single piece of the map is detected. The test covers the $K = 0$ channel, which equals the $K \to 0$ limit by continuity of the kernels.

Spatial spectrum of the temperature. Writing the $\ell = 2$ part as $Q_{ij}\hat{n}^i\hat{n}^j$,

$$P_{Q_{ij}Q_{ij}}(K \to 0) = \frac{3\epsilon^4}{8\pi^2} \int q^2 \mathrm{d}q \, P^2_{\zeta_1}(q) \, \mathcal{Q}_2^2(qt_\star, qt_o, qt_I) > 0. \tag{16}$$

This is white noise in the observer-label field of the measured quadrupole. It is not an angular spectrum. It says that quadrupoles measured by distant observers have a K^0 spectrum, and C_2 receives this region only with phase-space weight $\propto \int K^2 \mathrm{d}K$.

What is and is not curvature here. A constant-in-time long S_{ij} is a coordinate rescaling and is invisible in Θ, which sees only ΔS. That removal of the exactly homogeneous, time-independent part does not make the connected spectrum vanish, because the stress makes S_{ij} time dependent. The adiabatic expectation $\Theta_2 \sim K^2 t^2 \mathcal{R}$ fails because this $\mathcal{R}$ is not an adiabatic long mode: it arrives with an order-K^0 shear. Both $\mathcal{R}$ and Θ read the same tensor, via $\hat{K}\hat{K}$ and $\hat{n}\hat{n}$ respectively.

A Status of the claims

Claim	Status
ζ conservation, identities (a)–(d)	Exact derivation
a, n_2, I, W, $F_2^{\mathcal{R}}$ limit, (4)	Exact symbolic check, negative controls
$K \to 0$ continuity of kernels	Proved (entire Green function); numerically confirmed
$\mathcal{Q}_2$ evaluation and all pieces	Exact formula; quadrature; forward test
Absence of further logarithms in $\mathcal{Q}_2$; form (15)	Numerical regression with physical argument
Long-time behaviour beyond $\epsilon qt \ll 1$; shocks	Not resolved
Generic non-plane pair with $K \neq 0$ in the nonlinear forward test	Not performed; the forward test is $K = 0$
Complete C_ℓ from this white noise; superhorizon-to-subhorizon entry of the output	Not computed

B Code and reproduction

All code is in `solution/followup/`; `run_followup.sh` runs everything (about 10 minutes, peak memory about 2.7 GB).

File	Purpose
`plane_reduction.py`, `k0_fields.py`	reduced equations from the Ricci formula; closed-form proofs
`check_chart_R.py`	off-shell scalar-curvature invariance of (4)
`check_followup.py`	negative controls, tensor limit, stress, helicity, omission tests
`map_k0.py`	complete $\mathcal{Q}_2$ and pieces, Table 1
`forward_plane.py`	exact PDE and exact rays against $\mathcal{Q}_2$
`finite_K.py`	finite-K limit and $P_{22}(K)/A_0$
`late_time.py`, `nonuniform.py`	regression for (15); amplitude nonuniformity

The forward solver takes f' at nonzero wavenumber from the momentum constraint: evolving (T) directly excites the $-K^2/3$ scalar instability excluded by the constraints.

A stand-alone check of the decisive fields (`minimal_check.py`), which needs only SymPy and prints zero residuals:

```
"""Stand-alone: the K->0 pair fields solve the plane-symmetric equations (q=1, x=t/sqrt3)."""

# fmt: off
import sympy as sp

t, y, e, z1, z2 = sp.symbols("t y e z1 z2", positive=True)
x = t / sp.sqrt(3)
j0, c = sp.sin(x) / x, sp.cos(x)
C2 = sp.Function("C2")(t)                       # Cin(2x); C2'(t) = (1-cos 2x)/t
n1, h1 = c - j0, 6 * (1 + c - 2 * j0)
n2 = -sp.cos(2 * x) - 6 * c + 6 * j0 + 3 * sp.sin(2 * x) / x - 5 * j0**2
a2 = -78 - 36 * c - 6 * sp.cos(2 * x) + 72 * j0 + 36 * sp.sin(2 * x) / x - 24 * j0**2 + 24 * C2
g = z1 * sp.exp(sp.I * y) + z2 * sp.exp(-sp.I * y)
Z = e * g
f = e * h1 * g + e**2 * a2 * z1 * z2             # H = f diag(2/3,-1/3,-1/3); gamma = e^{2Z} exp H
nu = e * n1 * g + e**2 * n2 * z1 * z2
D = lambda u, v: sp.diff(u, v).subs(sp.Derivative(C2, t), (1 - sp.cos(2 * x)) / t)
fy, ny, zy = D(f, y), D(nu, y), D(Z, y)
```

```
ex = sp.exp(-2 * f / 3 - 2 * Z)
Ff = ex * (-fy**2 / 18 + fy * ny / 3 + fy * zy / 6 + D(fy, y) / 6 + 2 * ny**2
           + 4 * ny * zy - 2 * D(ny, y) + zy**2 - D(zy, y))
ft, nt = D(f, t), D(nu, t)
lap = ex * (D(ny, y) - (zy + fy / 3) * ny + 2 * (zy - fy / 6) * ny)
T_ = D(ft, t) + 2 * (1 / t + nt) * ft + 2 * Ff                                  # (T)
N_ = (3 * D(nt, t) - lap - 2 * ex * ny**2 + ft**2 / 6
      + 3 / t**2 * (sp.exp(-2 * nu) - 1))                                       # (N)
for name, eq in (("T", T_), ("N", N_)):
    first = sp.simplify(sp.diff(eq, e).subs(e, 0))
    second = sp.expand(sp.diff(eq, e, 2).subs(e, 0) / 2).coeff(z1, 1).coeff(z2, 1)
    print(name, "first:", first, "pair:", sp.simplify(sp.expand(sp.expand_trig(second))))
```

References

[1] K. Kohri and T. Terada, Phys. Rev. D 97 (2018) 123532.

[2] R.-G. Cai, S. Pi and M. Sasaki, Phys. Rev. D 102 (2020) 083528.